Practice Set 3

MA628: Financial Derivatives Pricing

February 9, 2026

---

1. The price, strike price, and time until expiration are given below for 3 European call options on the same non-dividend paying stock.

	Option Price	Strike Price	Expiration
Option A	\$8.00	\$50.00	1 year
Option B	\$7.70	\$52.00	1.5 years
Option C	\$7.50	\$53.00	2.0 years

An arbitrageur sees an arbitrage opportunity and therefore buys the 2-year option for \$7.50 and sells the 1.5-year option for \$7.70 at time 0. Subsequently, the actual stock prices emerge as described in the table below:

Time	Stock Price
1 year	\$50.00
1.5 years	\$52.50
2.0 years	\$52.50

The continuously compounded risk-free rate of return is 6%. Arbitrage profits are accumulated at the risk-free rate of return. Determine the value of the arbitrage profits at the end of 2 years.
Answer: 1.3091

2. Near market closing time on a given day, you lose access to stock prices, but some European call and put prices for a stock are available as follows:

Strike Price	Call Price	Put Price
\$45	\$12	\$4
\$55	\$7	\$9
\$60	\$4	\$12

All of the above options have the same expiration date. The risk-free interest rate is zero. After reviewing the information above, an arbitrager tells you that one could use the following zero-cost portfolio to obtain arbitrage profit: Short one put option with strike price 45; long 3 put options with strike price 55; lend \$1; and short some number of put options with strike price 60. Is arbitrager correct ? Justify your answer.